

Introduction

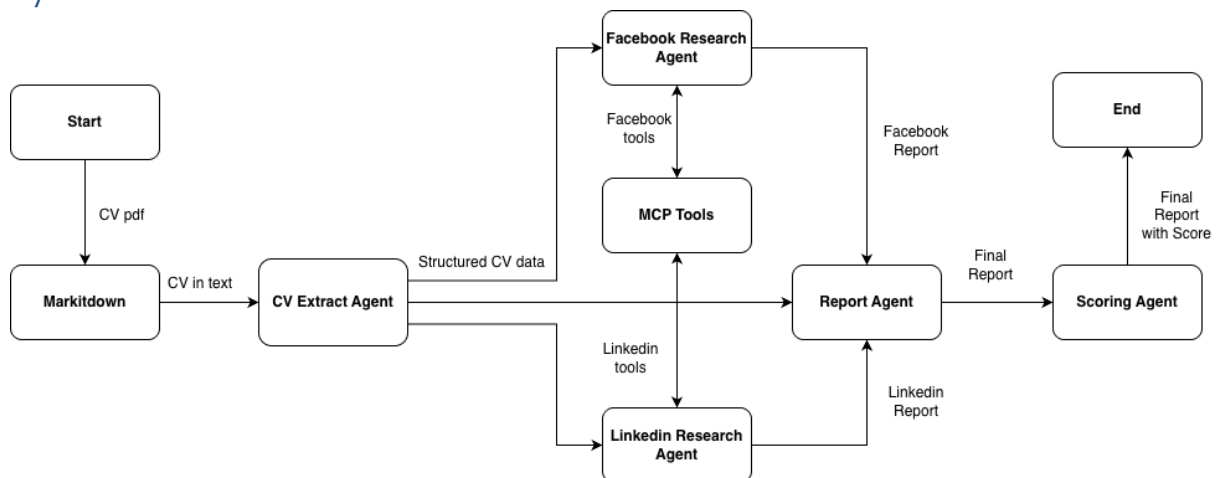
This assignment implements an AI agent system that automates CV verification against social media profiles (LinkedIn and Facebook) for Know Your Customer (KYC) and recruitment compliance processes. The system addresses the manual verification challenge by:

1. Extracting structured information from CV documents
2. Detecting discrepancies and timeline inconsistencies
3. Cross-referencing candidate claims against LinkedIn and Facebook profiles via MCP servers
4. Generating a comprehensive verification report with a trustworthiness score (0-100)

This solution uses Gemini 2.0 Flash (via Vertex AI) as the LLM for information extraction and reasoning. The system leverages the MCP server provided by instructor for LinkedIn and Facebook profile access.

This report documents the system architecture, agent workflow design, tool usage strategy, and sample verification results.

System Architecture



The solution follows a converter plus 5-agent pipeline with clear separation of concerns:

1. **Markdown Converter:** Converts PDF CV → structured markdown text
2. **CV Extract Agent:** Markdown CV → Structured CV data to be verified
3. **Facebook Research Agent:** Structured CV + Facebook MCP tools → Facebook Verification Report
4. **LinkedIn Research Agent:** Structured CV + LinkedIn MCP tools → LinkedIn Verification Report
5. **Report Agent:** Struct CV + FB Report + LI Report → Final Report
6. **Scoring Agent:** Final Report → numeric score (0-100 → 0-1)

Design Decisions

1. Hybrid LLM + Code Approach for Internal Inconsistency

Problem: LLMs miss subtle internal inconsistency, with only around 70% accuracy

Solution: this design uses LLM extracts structured data, then uses Python code validates with strict rules:

1. Timeline overlap, e.g., 2021-2024 Job A, 219 – 2025 Job B
2. Future date
3. Start date > end date
4. Multiple present job

It successfully obtained 100% overlap detection accuracy

2. Either/Or Matching Logic (Not Both Required)

Problem: Requiring both platforms penalize legitimate cases where one profile is incomplete or missing, while either one profile is enough to support the claims in CVs.

Solution: Clearly stated in the prompt - Support from either platform = valid evidence

- CV="BCG Engineer" + LinkedIn="BCG Engineer" = MATCH (Facebook irrelevant)
- Only flag when both platforms actively contradict CV

3. Strict Mismatch Hierarchy

MAJOR (LOW confidence):

- Company mismatch (EY vs PwC)
- Education mismatch (PhD vs MSc)
- Different schools

MINOR:

- Slightly different title (Senior Engineer vs Engineer)
- Unclear degree (BSc vs BSc in Engineering)
- 1 year date discrepancy

Agent Workflow and Tool Usage Strategy

Phase 1: Information Extraction

- **LLM Task:** Parse CV fields (name, education, experience, skills)
- **Code Task:** Detect overlaps using interval intersection algorithm
- **Output:** JSON with internal_inconsistent array

Phase 2: Social Media Verification

Facebook Agent Workflow:

1. search_facebook_users(q="John Smith", limit=20, fuzzy=True)
2. Filter by city from output results (not input parameter)
3. get_facebook_profile(userid) until good match found
4. Apply mismatch rules (company mismatch = MAJOR, PhD vs MSc = MAJOR, Senior vs regular = MINOR)
5. Generate markdown with Match Confidence and fields breakdown

LinkedIn Agent Workflow:

1. search_linkedin_people(q="John Smith", location="Singapore") for EVERY location
 - Search city BEFORE country
 - DO NOT combine "Tokyo, Japan" - separate calls
2. Analyze headlines to pick best matches where headline/industry/location match CV title
3. get_linkedin_profile(person_id) to get profile details
4. Verify: Company Names, Job Titles, Dates, Education (University, Degree)
5. Apply mismatch rules:
 - PhD vs MSc/Master's = major + LOW/MEDIUM confidence
 - Different schools = major
 - Unclear degree (BSc vs BSc in Engineering) = minor
 - Senior vs regular title = minor
 - Different field/industry = major
 - Exact company match = strong, no match = uncertain (NOT matched)

Phase 3: Report Synthesis**Report Agent Tasks:**

1. Synthesize clear verification report for human reviewer
2. Highlight overall alignment between CV and social profiles
3. Identify key matches that support candidate's claims
4. Flag key discrepancies with severity assessment
5. Note key missing education or experience in CV but missing on BOTH LinkedIn and Facebook
6. Identify areas of uncertainty (missing or ambiguous data)

Key Principle:

- SUPPORT FROM EITHER PLATFORM = MATCH (not discrepancy)
- Only include discrepancies between CV and BOTH social profiles (missing or mismatch)
- DO NOT include discrepancies between Facebook and LinkedIn profiles
- IF CV is supported by either Facebook or LinkedIn, it is in MATCH ONLY
 - CV="Engineer" + LinkedIn="Engineer" = MATCH (ignore Facebook)
 - CV="BCG" + Facebook="BCG" = MATCH (ignore LinkedIn)

Red Flags:

- Compare Company Names, Job Titles, Dates
- Compare Education (University, Degree)
- CV's experience not found on social profiles → major AND put in "Key Missing"
- Small date discrepancies (1 year off) = minor issues
- Major discrepancies (never worked there, wrong degree) = major red flags
- Internal consistency: check if social profile clarifies

Scoring Agent:

Produces final numeric score 0-100, and then scale down to 0-1 for evaluation, with scoring Scale:

- 100: Perfect match
- 75-99: Minor discrepancies (1yr date off, similar titles)
- 50-74: Ambiguous matches with multiple minor discrepancies or internal inconsistencies
- 35-49: Major discrepancies or major internal inconsistencies not resolved by social media
- 20-34: Many major discrepancies or major internal inconsistencies
- 0-19: Clear fabrications or no verifiable profile found

Testing Result

The system is tested towards 5 provided CVs.

1. CV_1.pdf

Name: John Smith

Report Summary: The CV of John Smith aligns well with his LinkedIn profile, showing consistency in his name, location, current title, education, and experience at ByteDance. While multiple John Smith profiles exist on Facebook with differing job titles, the LinkedIn profile provides strong support for the information presented in the CV. Overall, the verification process indicates a low risk.

Final Score: 0.85 (PASS)

Ground Truth: Pass

Correctness: Yes

2. CV_2.pdf

Name: Minh Pham

Report Summary: The CV of Minh Pham shows good alignment with his LinkedIn profile, particularly regarding his work experience at BCG and Tencent, as well as his education at The University of Hong Kong. There are discrepancies with the Facebook profile regarding current job title, education, and current company. Overall, the LinkedIn profile supports the CV's claims, while Facebook presents conflicting information.

Final Score: 0.75 (PASS)

Ground Truth: PASS

Correctness: Yes

3. CV_3.pdf

Name: Wei Zhang

Report Summary: The CV of Wei Zhang shows good alignment with their Facebook and LinkedIn profiles. Key details such as name, current role as Engineer at PwC, and education at the University of Tokyo are consistent across all platforms. There are no significant discrepancies, indicating a reliable profile.

Final Score: 0.95 (PASS)

Ground Truth: PASS

Correctness: Yes

4. CV_4.pdf

Name: Rahul Sharma

Report Summary: The CV of Rahul Sharma presents several inconsistencies when compared to his Facebook and LinkedIn profiles. While the name matches across all platforms, discrepancies arise in his current title, education, and work experience. The CV indicates a 'Legal Professional' role, a PhD from Tsinghua University, and experience at Microsoft and StartupXYZ. However, Facebook shows a 'Scientist' at BCG with a Master's degree, while LinkedIn shows a similar experience at Microsoft but with different job titles and dates. The absence of StartupXYZ in both social profiles and the conflicting details raise concerns about the accuracy of the CV.

Final Score: 0.35 (REJECTED)

Ground Truth: REJECTED

Correctness: Yes

5. CV_5.pdf

Name: Rahul Sharma

Report Summary: The CV of Rahul Sharma presents several inconsistencies when compared to his Facebook and LinkedIn profiles. While the name matches across all platforms, significant discrepancies arise in education, current job title, and employment history. The CV also contains internal inconsistencies regarding overlapping employment dates. These discrepancies raise concerns about the accuracy and reliability of the information presented in the CV.

Final Score: 0.35 (REJECTED)

Ground Truth: REJECTED

Correctness: Yes

Conclusion

In conclusion, this project delivers a multi-agent CV verification system that reliably cross-references candidate claims against LinkedIn and Facebook profiles via MCP servers. On the 5 sample CVs tested, the system achieves 100% accuracy in timeline overlap detection, discrepancy classification, and trustworthiness scoring.

GitHub Repository

The full implementation and this report are available

at: <https://github.com/TommyS725/FTEC5660/tree/main/homeworks/hw2>